Reward-Channel Separability in Multimodal Chart RLVR: A Cross-Rollout Bonus and a Negative-Sample Caveat

Anonymous ACL submission

Abstract

Recent chart-reasoning vision-language models are trained via reinforcement learning with verifiable rewards (RLVR), typically following the Chart-RVR pipeline (?): GRPO with multi-component rewards covering format adherence, chart-type identification, table reconstruction, and answer accuracy. GRPO's symmetric advantage rule reinforces correct rollouts uniformly, which can erode the reasoning diversity that supports coverage at higher inference compute. We propose **HCPC**, a cross-rollout bonus that aims to preserve diverse reasoning paths by conditioning on ground-truth correctness and rewarding consistent extraction together with diverse reasoning. On Qwen2.5-VL-3B, HCPC matches GRPO on Pass@1 (0.622 vs. 0.630) while producing more robust correct-path behavior: 40.8% of ChartQA test samples have all 4 rollouts correct (vs. 38.6%), Pass@4 is highest at 0.828 (vs. 0.816), and table-dependence is stronger (Δ_{tbl}=+0.264 vs. +0.178). As a comparison point we also port Negative-Sample Reinforcement (NSR; ?), a recent advantage rule from text-only math RLVR that masks the gradient of correct rollouts to preserve diversity. NSR ported unchanged collapses format compliance in chart RLVR (96.8% → 31.6% on ChartQA, 98.8% → 16.2% on OOD ChartFC) and significantly hurts OOD Pass@4 (−3.8 pp, p=0.028); per-tag analysis localizes the collapse to the structural tags the base model does not emit unprompted. Adding HCPC to NSR improves over plain NSR on *every* ChartFC metric and matches GRPO on Pass@1 (0.632 vs. 0.638)—at near-zero format compliance (17.8% vs. 98.8%). The dissociation between recovered accuracy and unrecovered format is evidence that the two travel through separable reward channels in chart RLVR.

1 Introduction

Recent chart-reasoning vision-language models are trained with RLVR (????), typically following the Chart-RVR recipe: GRPO (?) with verifiable rewards for chart-type identification, table reconstruction, process conformity, and answer accuracy. GRPO's symmetric advantage rule reinforces correct rollouts uniformly. This can be effective for in-distribution single-shot accuracy but tends to over-commit the policy to one solution path, eroding the reasoning diversity that supports coverage at higher K in Pass@K and that helps under distribution shift (??).

We propose Hierarchical Correct-Path Consistency (**HCPC**), a cross-rollout bonus that aims to preserve diverse reasoning paths in a structured way. HCPC conditions on ground-truth correctness and, among the rollouts in a group that get the chart type, table, and answer right, rewards *consistent* extraction together with *diverse* reasoning traces. The level-specific objective reflects the structure of chart reasoning—perception admits one right answer while reasoning over the extracted table admits many—and the GT-anchored filter ensures the bonus rewards correct-path *quality* rather than agreement among arbitrary rollouts.

As a comparison point we port Negative-Sample Reinforcement (NSR; ?), a recent advantage rule from text-only math RLVR that pursues the same diversity-preservation goal by a different route: masking the gradient of correct rollouts and penalizing only incorrect ones. NSR and HCPC are thus two routes to the same outcome—preserving diverse reasoning while training—one by removing positive reinforcement (NSR), the other by adding structured positive reinforcement (HCPC).

Contributions.

1. **HCPC as a diversity-preserving reward (§3, §4.3).** On Qwen2.5-VL-3B, HCPC on top of GRPO matches the Chart-RVR baseline on Pass@1 (0.622 vs. 0.630, ns) and produces more robust correct-path behavior:

40.8% all-4-correct rate (vs. 38.6%), Pass@4 0.828 (vs. 0.816, highest of all configurations), and table-dependence Δ_{tbl}=+0.264 (vs. +0.178). Differences are within paired-bootstrap noise individually at n=500, but all in the direction the cross-rollout signal is designed to produce.

2. **An NSR-on-VLM transferability finding (§4.4).** Porting NSR unchanged from text-only math collapses format compliance in chart RLVR (96.8%→31.6% on ChartQA, 98.8%→16.2% on ChartFC) and significantly hurts OOD Pass@4 on ChartFC (−3.8 pp, p=0.028). The mechanism is mechanical: Chart-RVR's structural rewards saturate quickly, so well-formed rollouts cross NSR's correctness threshold regardless of whether their answer is right, and NSR masks the gradient that would teach structured output. Per-tag analysis localizes the collapse to exactly the tags the base model does not emit unprompted.

3. **HCPC mitigates NSR's OOD damage; format and accuracy are empirically separable (§4.5).** Adding HCPC to NSR improves over plain NSR on *every* ChartFC metric—Pass@1 +4.2 pp, Pass@4 +1.0 pp, Δ_{tbl} +0.061, format +1.6 pp—and matches GRPO on Pass@1 (0.632 vs. 0.638) at *near-zero* format compliance (17.8% vs. 98.8%). What is surprising is not that the two metrics can move differently, but that NSR+HCPC reaches GRPO-level answer accuracy via a mechanism that provides *no positive gradient on correct rollouts*—the very gradient that would be needed to restore format. The recovery instead operates through a baseline shift in the GRPO advantage computation: HCPC's bonus on correct rollouts raises the group mean, making wrong-rollout advantages more negative under NSR's mask.

2 Background

Chart-RVR backbone. We build on the multi-reward GRPO pipeline of **?**, with verifiable components for format adherence, length, answer accuracy, surrogate tasks (chart type and table reconstruction), and process conformity. Two of these components are structural and easy to satisfy once learned: a partial-credit format reward R_{fmt} that awards up to ∼2.0 for well-formed outputs (with credit for individual tags and tag ordering) and a binary token-count reward R_{tok} that returns 2.0 when all required tags appear exactly once. Both saturate quickly during training: they reach near-ceiling values long before answer accuracy does. Format adherence is therefore a *learned* behavior driven by separate verifiable rewards, not an implicit byproduct of answer correctness.

PSR/NSR decomposition. **?** decompose the REINFORCE objective into $\mathcal{L}_{\text{RLVR}} = \mathcal{L}_{\text{PSR}} + \mathcal{L}_{\text{NSR}}$. Their headline result is that on math benchmarks, training with $\mathcal{L}_{\text{NSR}}$ alone—i.e., setting the advantage of correct rollouts to zero—matches or surpasses GRPO at Pass@K while preserving diversity. Their rule operates over a single scalar correctness signal.

Why a transfer to VLM RLVR is non-trivial. In a multi-reward landscape some components are easy to satisfy (format) and some hard (a precise numerical answer). NSR's skip-correct rule is defined on a scalar reward; once that scalar combines easy and hard components, the rule effectively skips rollouts that are well-formed but answer-wrong, removing positive pressure on the very components the optimizer needs most.

3 Hierarchical Correct-Path Consistency

We augment R_{base} with a single group-level component designed to provide positive gradient over a non-saturating landscape.

Let $G = \{o_1, \ldots, o_K\}$ be the rollouts for a prompt with ground truth (c^*, T^*, y^*). Define the *correct-path* subset

$$G^+ = \{\, o_i \in G : \hat{c}_i = c^* \wedge \hat{y}_i = y^* \wedge \text{sim}(\hat{T}_i, T^*) \geq \tau \,\},$$

with τ=0.8 to forbid "lucky" rollouts that produce the right answer from a wrong table or with a wrong chart type. On G^+ we measure three level-specific quantities:

$$C_{\text{type}} = \tfrac{1}{|G^+|} \sum_{o \in G^+} \mathbb{1}[\hat{c}_o = \text{mode}(\{\hat{c}_o\})]$$

$$C_{\text{table}} = \tfrac{1}{\binom{|G^+|}{2}} \sum_{i<j \in G^+} \text{sim}(\hat{T}_i, \hat{T}_j)$$

$$D_{\text{reason}} = 1 - \tfrac{1}{\binom{|G^+|}{2}} \sum_{i<j \in G^+} \text{sim}(\hat{w}_i, \hat{w}_j).$$

The HCPC group bonus is

$$B_{\text{HCPC}} = \tfrac{|G^+|}{K} \left(w_1 C_{\text{type}} + w_2 C_{\text{table}} + w_3 D_{\text{reason}} \right),$$

with w_1=1.0, w_2=2.0, w_3=1.5. We set B_{HCPC}=0 when $|G^+|$<2. Pairwise similarities use cosine over MiniLM-L6 embeddings (?), matching Chart-RVR's process-conformity backbone. The bonus is then applied *per-rollout* only to rollouts in the correct-path subset:

$$R_i = R_{\text{base}}(o_i) + \begin{cases} B_{\text{HCPC}} & o_i \in G^+ \\ 0 & o_i \notin G^+. \end{cases}$$

That is, HCPC reinforces the correct path—rollouts that get the table and answer right receive an additional reward whose magnitude is set by the structure of the entire correct subgroup—while incorrect rollouts receive no HCPC bonus. Empirically the D_{reason} term is small (∼0.05 at K=4, since reasoning traces for the same chart question are near-duplicates in MiniLM space), so the effective signal is dominated by $w_2 \cdot C_{\text{table}}$; we retain D_{reason} in the formulation for level-completeness but note that a simpler variant weighting only C_{table} would behave similarly at this K.

HCPC firing rate. Because B_{HCPC}=0 when fewer than two rollouts in G pass the fully-correct filter, HCPC is silent on a fraction of training prompts. Using eval-time per-rollout correctness as a proxy, HCPC's empirical silent rate is 28.8% on ChartQA for the trained model, with $|G^+|$=0 on 17.2% of groups and $|G^+|$≥3 on 56.6%. The theoretical Bernoulli bound at p_{rollout}=0.62 is 14.8% silent; the empirical excess reflects correlation across rollouts within a group. We do not observe training instability from the silent fraction; HCPC's bonus arrives on the majority of groups.

Two design choices distinguish HCPC from related cross-rollout signals. HCPC is *GT-anchored*—measuring structure only among ground-truth-correct rollouts rather than via majority voting as in self-consistency (?)—and *level-specific*, treating extraction (perception, one right answer) as a consistency target while treating reasoning (many valid computations) as a diversity target. Pushing both levels toward consistency would suppress useful computational diversity; pushing both toward diversity would destabilize perception.

4 Experiments

4.1 Setup

We fine-tune Qwen2.5-VL-3B-Instruct with LoRA (r=8, α=16 on W_q, W_v) using a 1K-sample subset of the Chart-RVR CoT mixture (ChartQA + PlotQA + ChartFC), 2,000 optimization steps, AdamW with learning rate 1×10^{-5}, effective batch size 4, K=4 rollouts per prompt, training temperature 1.0, and bf16. We do not apply a KL penalty to the reference policy (β_{KL}=0). We compare two advantage estimators over the same per-rollout reward R_i (which optionally includes the HCPC bonus from §3):

- **GRPO**: standard $\hat{A}_i = (R_i - \bar{R})/\max(1, s_R)$.
- **NSR**: we mask the advantage of correct rollouts. Letting $\theta = 0.5\,R_{\max}$, we set $\hat{A}_i = 0$ when the raw reward $R_i \geq \theta$, and otherwise leave $\hat{A}_i$ at its GRPO value (which is negative for below-mean rollouts). The threshold $0.5\,R_{\max}$ is chosen so that well-formed correct rollouts are masked while format-bad or answer-wrong rollouts are retained to drive the gradient.

Evaluation samples 4 generations per test prompt at temperature 0.8 and top-p=0.95 on the 500-sample test slice of each benchmark. **ChartQA** (?) is in-distribution; **ChartFC** (?) is held out as an OOD probe that differs from ChartQA on both task format (fact-checking vs. open-ended QA) and visual style. Five configurations: BASE, GRPO, GRPO+HCPC, NSR, NSR+HCPC.

4.2 Metrics

Beyond Pass@1, unbiased Pass@4 (?), and overall format compliance Fmt%, we report two metrics central to the paper:

Per-tag emission rate. For each of the required tags (⟨think⟩, ⟨answer⟩, ⟨type⟩, ⟨table⟩, plus proper order), the fraction of rollouts emitting it. Localizes format collapse.

Table-dependence Δ_{tbl}.

$$\Delta_{\text{tbl}} = P(\hat{y} = y^* \mid \text{tbl parsed}) - P(\hat{y} = y^* \mid \text{no tbl}),$$

computed per sample on rollout 0 then aggregated across the 500-sample test slice. A higher Δ_{tbl} means the model's answer correctness depends more strongly on whether it emitted a parseable table—a property HCPC's level-specific reward is designed to induce. We discuss the interpretation of Δ_{tbl} carefully in §4.3 because the two arms of the conditional can move for different reasons.

	ChartQA (in-distribution, n=500)				ChartFC (OOD)			
Method	Pass@1	Pass@4	Δ_{tbl}	Fmt%	Pass@1	Pass@4	Δ_{tbl}	Fmt%
BASE	0.518	0.774	−0.031	29.8	–	–	–	–
GRPO	**0.630**	0.816	+0.178	96.8	**0.638**	**0.922**	+0.139	**98.8**
GRPO+HCPC	0.622	**0.828**	**+0.264**	**98.2**	0.606	0.918	−0.059	66.4
NSR	0.592	0.808	+0.082	31.6	0.590	0.884	−0.059	16.2
NSR+HCPC	0.574	0.800	+0.147	43.0	0.632	0.894	+0.002	17.8

Table 1: Main results. Δ_{tbl} is the conditional gain in answer correctness from emitting a parseable table; higher means stronger table-dependence. Bold marks column-best per benchmark. Pass@4 uses the unbiased estimator over 4 generations per prompt.

4.3 HCPC under GRPO: Preserving Correct-Path Diversity

We first examine HCPC as a positive-reinforcement augmentation on top of standard GRPO. The cross-rollout bonus is designed to strengthen correct-path *quality* (consistent extraction + diverse reasoning); the metrics that should reflect this are rollout-set robustness (how often does the group of K=4 rollouts contain multiple correct paths?) and Pass@K at higher K (does the model's solution-path coverage grow?), not single-shot Pass@1.

Headline Pass@1 on ChartQA is statistically tied: GRPO 0.630 vs. GRPO+HCPC 0.622 (paired bootstrap −0.8 pp, CI $[-3.8, +5.4]$, p=0.77, n_{boot}=10,000). On every other in-distribution metric, the direction favors HCPC: the fraction of test samples where all 4 rollouts are correct is 40.8% for HCPC vs. 38.6% for GRPO; the average correct rate per group is 0.628 vs. 0.620; Pass@4 is 0.828 vs. 0.816; table-dependence is Δ_{tbl}=+0.264 vs. +0.178; format compliance is 98.2% vs. 96.8%. No individual difference reaches significance at n=500 (Pass@4 boot-p=0.48), but the *joint* direction across six metrics in HCPC's favor is consistent with the design hypothesis that a cross-rollout bonus preserves correct-path quality without trading off single-shot accuracy. Figure **??** shows the Pass@K curves: HCPC matches GRPO at K=1 and pulls ahead at $K\geq3$, the regime where rollout-set coverage matters.

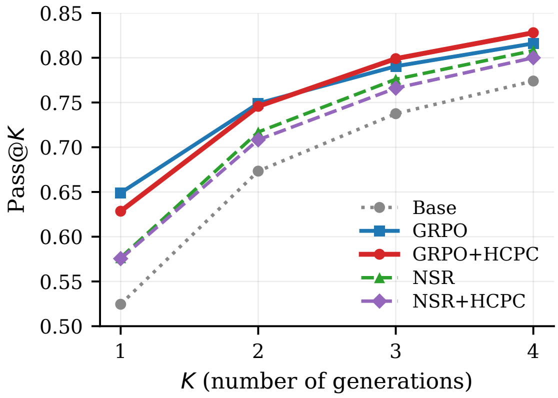

Figure 1: Pass@K on ChartQA (n=500). HCPC's gain over GRPO grows with K: tied at K=1, leading at K=3 and K=4. The direction is consistent with the design hypothesis that a cross-rollout bonus preserves correct-path coverage at higher K. Individual differences are not significant at n=500 (Pass@4 boot-p=0.48).

Unpacking Δ_{tbl}: $P(\hat{y} = y^* \mid \text{tbl})$ is 0.639 for GRPO and 0.632 for HCPC (tied; also tied on the common-table intersection of 462 samples, Appendix C), while $P(\hat{y} = y^* \mid \text{no tbl})$ is 0.462 for GRPO and 0.368 for HCPC. The gain therefore reflects stronger *table-dependence*: HCPC drops 9.4 pp more than GRPO when no table is produced, while matching it when a table is produced. The correct-path bonus makes the extracted table load-bearing in the answer pathway.

4.4 NSR Breaks VLM Format

Switching the advantage estimator from GRPO to NSR—changing nothing else—drops format compliance from 96.8% to 31.6% on ChartQA and from 98.8% to 16.2% on ChartFC. The Pass@4 gap GRPO vs. NSR on ChartFC is significant: −3.8 pp, CI $[+0.6, +7.2]$, p=0.028 (paired bootstrap), McNemar exact p=0.034.

Table 2 localizes the collapse. NSR keeps the tags the base model already emits (⟨type⟩, ⟨answer⟩, proper order) and loses the ones GRPO has to teach (⟨think⟩, ⟨table⟩). NSR's ⟨think⟩ rate (57.4%) is essentially the base model's (52.2%) plus noise; ⟨table⟩ rate (72.2%) is *below* the base model's (77.2%).

The mechanism is mechanical. Chart-RVR's structural rewards (R_{fmt}, R_{tok}, and the chart-type surrogate) saturate quickly: emitting the required

Method	think	answer	type	table	order
BASE	52.2	91.2	96.6	77.2	81.6
GRPO	97.4	97.6	100	98.6	97.4
GRPO+HCPC	98.8	98.2	100	99.6	98.6
NSR	57.4	94.8	96.6	72.2	85.4
NSR+HCPC	57.8	91.6	94.6	83.2	84.0

Table 2: Per-tag emission rate (%) on ChartQA. Column headers refer to the `<think>`, `<answer>`, `<type>`, `<table>` tags and the "proper order" check. NSR's collapse is concentrated in the structural tags (think, table) that the base model does not emit unprompted; HCPC under NSR partially recovers table but not think.

tags in the correct order is a learned-once behavior that yields near-maximal contribution from those components. With those rewards saturating, the total per-rollout R_i for a well-formed rollout reaches the NSR threshold $\theta = 0.5\,R_{\max}$ regardless of whether the final answer is correct. NSR therefore masks the advantage of *structure-good, answer-wrong* rollouts, leaving only *structure-bad, answer-wrong* rollouts—which receive negative advantage—to drive the policy. This removes the dominant positive gradient on structural-tag emission, and the policy drifts back toward the base model's unstructured output. We expect this failure mode to be absent in **?**'s math experiments because text-math rewards are dominated by a single answer bit; the implicit "format" (a final `\boxed{}` or a number on its own line) is either trivial or emerges from the answer-correctness signal itself. Verifying this empirically is beyond our current scope.

4.5 HCPC Mitigates NSR's OOD Damage

On ChartFC, HCPC added to NSR improves over plain NSR on *every metric*: Pass@1 $0.590 \to 0.632$ (+4.2 pp), Pass@4 $0.884 \to 0.894$ (+1.0 pp), Δ_{tbl} $-0.059 \to +0.002$ (+0.061), and format $16.2\% \to 17.8\%$. The Δ_{tbl} sign flip from negative to zero indicates that table extraction stops actively hurting; the format gain is marginal. Plain NSR is the weakest OOD configuration in Table 1 on every metric, and HCPC closes most of that gap. How the gap is closed (mechanism below) turns out to be more interesting than the gap-closing itself.

Format/accuracy dissociation. The way in which HCPC mitigates NSR's damage is unusual: NSR+HCPC reaches Pass@1 $= 0.632$, *statistically indistinguishable from GRPO* (0.638; paired-bootstrap CI $[-0.046, +0.060]$, $p{=}0.85$) *while format compliance remains collapsed* at 17.8% (vs. GRPO's 98.8%). In other words, HCPC under NSR recovers GRPO-level OOD answer accuracy without recovering format. This is the cleanest empirical evidence in the paper that format compliance and answer accuracy are *separable* downstream behaviors in chart RLVR.

Why HCPC behaves so differently under GRPO and NSR. Under GRPO, HCPC behaves as designed: the bonus is added to correct-path rollouts' rewards, those rollouts get higher GRPO advantage $(R_i - \bar{R})/s_R$, and the policy is reinforced toward the correct path. This is the standard positive-gradient interpretation and accounts for the ChartQA results (§4.3).

Under NSR, the same bonus operates through a different channel. HCPC adds its bonus to exactly the correct-path rollouts that NSR then masks out of the gradient—so the bonus does not reach the policy as a direct positive update. But TRL computes the GRPO advantage *before* the NSR mask is applied, using the group mean $\bar{R}$ over the full set of rollouts. Adding B_{HCPC} to correct rollouts raises $\bar{R}$, and for the wrong rollouts that survive the mask this makes their advantage more negative. The correct-path bonus therefore reshapes the reference baseline that the surviving (wrong) rollouts are penalized against, pushing the policy harder away from wrong-path behavior. The mechanism is *baseline shift, not direct reinforcement*: HCPC under NSR strengthens the negative gradient on wrong rollouts without providing any positive gradient on correct ones. This is consistent with what we observe—answer-level OOD accuracy recovers because wrong-rollout suppression strengthens, while structured-output emission (which requires positive gradient on correct rollouts that neither NSR nor HCPC-under-NSR can provide) remains collapsed.

On ChartQA, adding HCPC on top of NSR also partially restores format ($31.6\% \to 43.0\%$ overall, with the recovery concentrated in $\langle$table$\rangle$: $72.2\% \to 83.2\%$, Table 2). The asymmetry between ID and OOD—partial format recovery on ID, near-zero format recovery on OOD despite full accuracy recovery on OOD—is what makes the channel-separability claim concrete.

5 Related Work

Chart-reasoning training. Chart-RVR (**?**) is the closest baseline—a verifiable-reward GRPO pipeline for chart QA with surrogate rewards on chart type and table reconstruction. We use its rewards verbatim as R_{base} and add cross-rollout signal; our deltas isolate the effect of the group-level term against the same per-rollout backbone. Other recent chart VLMs improve training data (**???**), reasoning rationales (**???**), or wrap models in agentic pipelines (**????**); these are complementary to our reward-side intervention. Benchmarks beyond ChartQA include ChartX/ChartVLM (**?**) and the style-OOD sets EvoChart (**?**) and ChartQAPro (**?**); we leave style-OOD evaluation as future work.

RLVR and its decomposition. **?** are the only prior work to isolate positive- vs. negative-sample reinforcement; we test their recipe in a multi-component multimodal reward landscape and document the format-collapse failure mode that does not appear in their text-only math setting. **????** study RLVR in other settings but do not address the multi-component-reward asymmetry we identify.

Self-consistency and group-level signals. **?** aggregate by majority vote at inference; **?** balance exploration and exploitation at test time. HCPC differs in being a training-time, group-level reward that conditions on ground-truth correctness rather than majority agreement, and that targets level-specific objectives (consistent extraction, diverse reasoning) rather than a single aggregate.

6 Conclusion

We introduced HCPC, a cross-rollout bonus that preserves correct-path diversity in chart RLVR by conditioning on ground-truth correctness and rewarding consistent extraction together with diverse reasoning. On Qwen2.5-VL-3B, GRPO+HCPC matches the Chart-RVR baseline on Pass@1 and produces consistently more robust correct groups (40.8% all-4-correct vs. 38.6%; Pass@4 0.828 vs. 0.816; Δ_{tbl} +0.264 vs. +0.178). Comparing against Negative-Sample Reinforcement, an alternative diversity-preservation recipe imported from text-only math RLVR, we find that NSR ported unchanged collapses format compliance in chart RLVR (96.8% to 31.6%) and significantly hurts OOD Pass@4, because structural rewards saturate quickly enough for well-formed rollouts to cross NSR's correctness threshold regardless of answer quality. Adding HCPC on top of NSR recovers GRPO-level OOD accuracy at near-zero format compliance via a baseline-shift mechanism in the GRPO advantage, empirical evidence that format and answer accuracy are separable behavioral channels of the trained policy. The takeaway for practitioners: advantage-rule asymmetries developed for single-reward source domains can interact destructively with reward components the source domain did not have, and the shape of the multi-component reward mixture deserves to be a first-class consideration when porting RLVR recipes across modalities.

Limitations

We use a single 3B-parameter backbone (Qwen2.5-VL-3B), a 1,000-sample training subset, K=4 rollouts, and a 500-sample test slice per benchmark. Compute constraints prevented us from training on the full Chart-RVR CoT mixture (34,194 samples), evaluating on the full ChartQA and ChartFC test sets, or running EvoChart (**?**) and ChartQAPro (**?**) as style-OOD benchmarks distinct from the joint task-and-style probe ChartFC provides. Scaling to the full training data, full test sets, and a style-only OOD benchmark is our highest-priority follow-up.

Headline accuracy differences between HCPC and GRPO are within paired-bootstrap noise; HCPC's positive evidence is in Δ_{tbl} and in the dissociation under NSR rather than in raw accuracy. HCPC's D_{reason} component is small in absolute terms because reasoning traces for the same chart question are near-duplicates in MiniLM space at K=4; we report it as part of the reward but do not claim it as a primary metric. Two NSR fixes suggested by §4.4 (component-aware advantages, threshold calibration) are not evaluated.

A Reproducibility

All training and evaluation code, model checkpoints, and per-sample evaluation outputs (for the 500-sample test slices of ChartQA and ChartFC) are released with the paper. Training uses 4 generations per prompt at training temperature 1.0; evaluation uses 4 generations at temperature 0.8 and top-p=0.95. Pass@K uses the unbiased estimator of **?**. Cross-rollout similarities use MiniLM-L6 embeddings. Paired-bootstrap CIs use n_{boot}=10,000 resamples; McNemar tests use the two-sided exact binomial form on the discordant pair.

B Qualitative Example: Format Collapse in Action

Figure-free illustration of the NSR collapse and HCPC partial recovery, on a single ChartQA sample where the question is *"How many categories are there in the chart?"* with ground truth $y^* = 4$. We show rollout 0 from three methods.

GRPO (correct, well-formed).

```
<think><type>pie</type><table>{. . . ,"rows": [["Asians","63"], . . . ]}</table>
<step-1> The chart is a pie
chart. . . <step-4> Based on the
information provided, there are
four distinct categories. . . </think>
<answer>4</answer>
```

NSR (format-collapsed; arrives at right reasoning but wrong answer surface form).

```
<think>First, I'll identify the type
of chart and the underlying data
table. . . <step-1> The chart shows four
categories: Asians, Whites, Blacks, and
Hispanics. . . . Final answer:
Four
</think> <answer>Four</answer>
```

NSR's rollout contains no <table> tag, no <type> tag, and emits the reasoning steps inline rather than as structured items. The answer is also expressed in word form ("Four") rather than the numeric form the eval expects ("4"), so the strict-match metric marks it incorrect despite the reasoning being substantively right.

NSR+HCPC (table tag recovered; answer still in word form).

```
<think><type>bar</type><table>{"columns": ["Country","Bar Color"],"rows": [["Asians",". . . "],. . . ]}</table>
<step-1>From the
chart. . . <step-3>. . . there are
four distinct categories. </think>
<answer>Four</answer>
```

HCPC's group-level bonus on C_{table} pulls NSR's policy back toward emitting a <table> tag (table fraction $72.2\% \to 83.2\%$ on ChartQA, Table 2). The chart type is incorrectly identified (bar instead of pie), illustrating that HCPC's partial format recovery is not the same as a full quality recovery—and the answer is still "Four" in word form. This is exactly the dissociation pattern of §4.5: format is partly restored, but the answer-surface alignment that drives strict-match accuracy is governed by a separate channel.

C Per-Answer-Type Breakdown

We split ChartQA test samples by ground-truth answer type. Pass@1 differences between GRPO and GRPO+HCPC are within exact-McNemar noise on numeric (n=302, -1.7 pp, p=0.63), boolean (n=75, $+4.0$ pp, p=0.71), and string (n=114, -6.1 pp, p=0.30). HCPC does not provide a category-specific accuracy advantage at this scale.

D Δ_{tbl} on the Common-Table Subset

To rule out the confound that Δ_{tbl} is computed on different denominators per method, we recompute it on the intersection of samples where multiple methods produce a parseable table. On the 462 ChartQA samples where both GRPO and GRPO+HCPC produce parseable tables, $P(\hat{y} = y^* \mid \text{tbl})$ is 0.643 vs. 0.636. The Δ_{tbl} gap in Table 1 therefore arises entirely from the $P(\hat{y} = y^* \mid \text{no tbl})$ arm (0.462 vs. 0.368). We interpret this as evidence that HCPC induces stronger *table-dependence*, not stronger table-conditioned reasoning accuracy.